

10.6.2

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Construct a pair of tangents to a circle of radius 4 cm from a point P at a distance of 7 cm from its centre O .

Solution:

The tangent directions $\mathbf{m}$ from an external point $\mathbf{h}$ to the circle

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{x} - r^2 = 0 \quad (1)$$

satisfy

$$\mathbf{m}^\top \Phi \mathbf{m} = 0 \quad (2)$$

where

$$\Phi = \mathbf{h}\mathbf{h}^\top - g(\mathbf{h})\mathbf{I} \quad (3)$$

With the point $\mathbf{h} = d\mathbf{e}_1$, we have $g(\mathbf{h}) = d^2 - r^2$. From (3),

$$\Phi = \begin{pmatrix} d^2 & 0 \\ 0 & 0 \end{pmatrix} - (d^2 - r^2) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (4)$$

$$= \begin{pmatrix} r^2 & 0 \\ 0 & -(d^2 - r^2) \end{pmatrix} \quad (5)$$

Since Φ is a diagonal matrix, its eigenvalues are:

$$\lambda_1 = r^2, \quad \lambda_2 = -(d^2 - r^2) \quad (6)$$

The matrix of orthonormal eigenvectors is

$$\mathbf{P} = \mathbf{I}$$

Hence, the unit direction vectors of the tangents are given by:

$$\mathbf{m} = \frac{1}{\sqrt{\lambda_1 - \lambda_2}} \mathbf{P} \begin{pmatrix} \sqrt{-\lambda_2} \\ \pm \sqrt{\lambda_1} \end{pmatrix} \quad (7)$$

From (6),

$$\lambda_1 - \lambda_2 = r^2 - (-(d^2 - r^2)) = d^2 \quad (8)$$

$$-\lambda_2 = d^2 - r^2 \quad (9)$$

Substituting into (7):

$$\mathbf{m} = \frac{1}{\sqrt{d^2}} \begin{pmatrix} \sqrt{d^2 - r^2} \\ \pm r \end{pmatrix} \quad (10)$$

$$= \frac{1}{d} \begin{pmatrix} \sqrt{d^2 - r^2} \\ \pm r \end{pmatrix} \quad (11)$$

The points of contact are given by

$$\mathbf{q} = \mathbf{h} + \kappa \mathbf{m} \quad (12)$$

Satisfies the circle equation

$$g(\mathbf{h} + \kappa \mathbf{m}) = (\mathbf{h} + \kappa \mathbf{m})^\top (\mathbf{h} + \kappa \mathbf{m}) - r^2 = 0 \quad (13)$$

$$\implies \kappa^2 (\mathbf{m}^\top \mathbf{m}) + 2\kappa (\mathbf{h}^\top \mathbf{m}) + g(\mathbf{h}) = 0 \quad (14)$$

For a tangent, κ has a single repeated root. Since $\mathbf{m}$ is a unit vector ($\mathbf{m}^\top \mathbf{m} = 1$):

$$\kappa = -(\mathbf{h}^\top \mathbf{m}) \quad (15)$$

$$\mathbf{h}^\top \mathbf{m} = (d\mathbf{e}_1)^\top \frac{1}{d} \begin{pmatrix} \sqrt{d^2 - r^2} \\ \pm r \end{pmatrix} = \frac{\sqrt{d^2 - r^2}}{d} \quad (16)$$

Hence,

$$\kappa = -\sqrt{d^2 - r^2}$$

The points of contact are:

$$\mathbf{q} = d\mathbf{e}_1 - \sqrt{d^2 - r^2} \frac{1}{d} \begin{pmatrix} \sqrt{d^2 - r^2} \\ \pm r \end{pmatrix} \quad (17)$$

$$= \begin{pmatrix} \frac{r^2}{d} \\ \mp \frac{r\sqrt{d^2 - r^2}}{d} \end{pmatrix} \quad (18)$$

Substituting $r = 4$, $d = 7$:

$$\sqrt{d^2 - r^2} = \sqrt{49 - 16} = \sqrt{33} \quad (19)$$

$$\mathbf{q} = \begin{pmatrix} \frac{16}{7} \\ \pm \frac{4\sqrt{33}}{7} \end{pmatrix} \quad (20)$$

Thus, the points of contact of the tangents are:

$$\boxed{\mathbf{q}_1 = \begin{pmatrix} \frac{16}{7} \\ \frac{4\sqrt{33}}{7} \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} \frac{16}{7} \\ -\frac{4\sqrt{33}}{7} \end{pmatrix}}$$

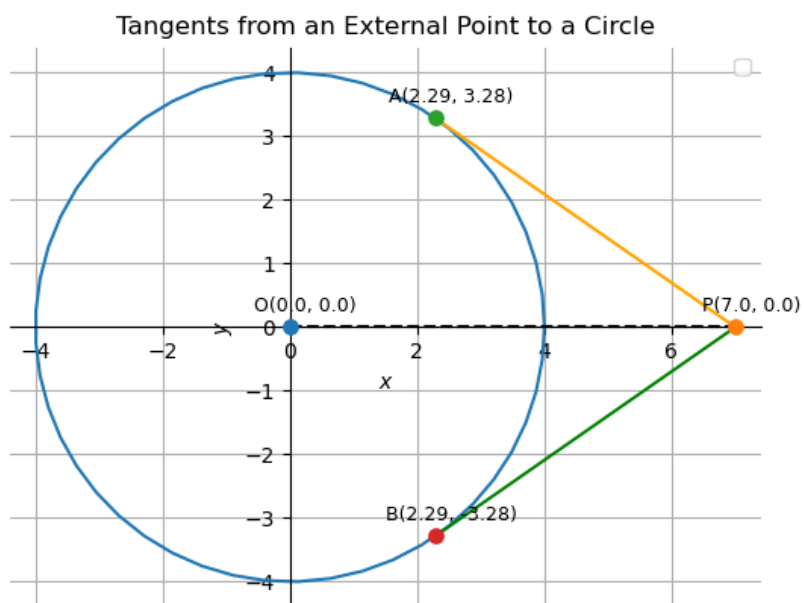


Figure 1